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Anytime Rule Compression and Rectified Logistic Modeling for Longitudinal Signals

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Summary: Interpretable models are essential in longitudinal sensor analytics where end-users must understand *why* a prediction was made. We present an anytime "logic-polishing" framework that compresses a sparse, L1-regularized logistic model of continuous features rectified (±1) and trained to produce a compact **m-of-K** rule list tuned to maximize Youden's **J**. On synthetic longitudinal datasets with known ground-truth rules, the approach reduces model complexity by up to ~50× (e.g., 344 non-zero coefficients → 7 rules at *N*=5,000; 349 → 7 at *N*=9,000; 257 → 8 at *N*=10,000) while matching or slightly improving J and maintaining AUC ≈ logistic baselines. AUC differences were negligible at the 0.1–0.3% level. A custom solver exploiting binarized inputs trained 1.5–2.3× faster than scikit-learn on the largest sets, and the polishing step added only fractions of a second. The method consistently achieved full adoption (no loss in J) under global K-policies.

Keywords: Interpretable machine learning, rule compression, longitudinal signals, L1 logistic regression, binarization, model distillation.

1. Introduction

In high-stakes sensor monitoring, stakeholders require transparent models with balanced sensitivity and specificity [1,2]. While L1-regularized logistic regression provides sparsity, even the resulting models can involve hundreds of small-magnitude weights that obscure the underlying decision logic for discretized longitudinal features. We address this gap with an **anytime rule compression** pipeline that (i) rectifies features to ±1 for direct logical semantics, (ii) trains an L1 logistic baseline (custom solver exploiting binary inputs), and (iii) "polishes" the model into a short **m-of-K** rule list by maximizing Youden's **J** at a chosen operating point.

1.1. Contributions

An anytime, J-preserving rule-compression algorithm that converts a rectified logistic model into a concise rule set; users can halt at any point to trade simplicity for accuracy. 2) A binarized (±1) feature set that yields human-readable atomic conditions and enables efficient coordinate-descent training. 3) Empirical evidence on synthetic longitudinal data with known ground truth: up to ~50× reduction in complexity with J preserved (or modestly improved) and AUC essentially unchanged for the complete set of relevant features. 4) Analysis of the vote magnitude **K** and threshold **k** (m-of-K) showing robustness across K∈{3,5,10} and consistent full adoption. 5) Speedups of ~1.5–2.3× vs. scikit-learn on large rectified models.

1.2. Approach

Our approach begins by transforming raw longitudinal sensor data into **rectified features** that enable logical interpretation. Given *M* sensor signals measured over time for each instance (e.g., a patient's longitudinal vital signs), we avoid relying on raw time-series values or handcrafted temporal features. Instead, we apply a simple rectification: each sensor reading is compared against a predefined threshold derived from its value at the time of a recorded event [3]. That is to say, the thresholds are derived from the data itself rather than supplied by an outside source, like a subject matter expert (SME). The resulting feature is encoded as +1 if the reading exceeds the threshold (high) and −1 if it falls below (low). For naturally binary or categorical sensors, this transformation simply preserves their form by mapping categories into the ±1 domain in a trivial transformation. This rectification provides a uniform binary input space, serving as the foundation for both efficient model training and the extraction of interpretable logical rules [4].

In addition, we use a mean-matching intercept for the rectified logistic model via the LASSO [5,6]:

$$\hat{b} = \log\left(\frac{p^*}{1-p^*}\right) - \mathbf{w}^\top \boldsymbol{\mu} \tag{1}$$

where $p^* = \frac{1}{N}\sum_{i=1}^{N} y_i$ is the empirical prevalence, $\boldsymbol{\mu}_j = \frac{1}{N}\sum_{i=1}^{N} X_{ij}$ are feature means for rectified inputs $X_{ij} \in \{-1, +1\}$, and $\mathbf{w}$ is the learned sparse coefficient vector. Eq. (1) anchors the model's logit at

the mean feature vector to the empirical class log-odds and integrates cleanly with the subsequent m-of-K rule construction.

1.3. Relationship to Prior Work

While the previous work [3] demonstrates that binarization itself is a powerful tool for reducing the obfuscating effect of multiple highly correlated features, and demonstrates this with several engineered (synthetic) data sets as well as complex real-world open source data sets, this short paper takes the additional step of collapsing the remaining small value irrelevant coefficients and presenting a clean, sparse function which accurately describes the logical relationship to the relevant input features.

This relationship comes at near zero cost if the relevant input features are related by a logical rule (e.g. a go/no-go relationship between combinations of input variables).

2. Method

Binarization of the input data and subsequent logistic regression fitting with L1-regularization (LASSO) is a crucial step in this method and makes the follow-on logical polishing possible. Notably, while this is not the only possible binarization method, this one is simple to implement and its successful usage has been empirically demonstrated in our previous work [3].

Algorithm 1 Binarization (adapted from [3])

Require: **X**, y (training set only)
Output: **A**, $r_j[b, c]$

1: Let $r_j[b, c]$ be the critical range for feature vector $\mathbf{X}_j$, where b is the minimum and c is maximum defining that range.

2: **procedure** BINARIZATION(**X**, y)

3: Let Z be the rows of **X** where y=1.

$Z \subseteq \mathbf{X}$, where $Z = \{\mathbf{X} \mid y = \mathbf{TRUE}\}$

4: Compute $r_{j=1..d} = [min(Z_j), max(Z_j)]$.

Note: If the min / max calculations induce brittleness in the solution (e.g. if there are outliers in the data set), using a robust quantile can provide a remedy.

5: Compute **A**, the set of transformed features, such that:

$a_{ij} = +1$ when $x_{ij} \geq r_j[b]$ and $x_{ij} \leq r_j[c]$

$a_{ij} = -1$ when $x_{ij} < r_j[b]$ or $x_{ij} > r_j[c]$

6: **end procedure**

***Important**: The critical ranges are only calculated using the training set, not the set-aside test set. The critical ranges are retained and used again when processing the test set. This prevents information leakage and maintains the integrity of the data.*

When binarization and model fitting are complete. The following algorithm implements the logical polishing step used to collapse the low magnitude coefficients after the LASSO fit of the binarized data.

Algorithm 2 Logical Polish

Require: $X \in \{-1, +1\}^{n \times d}$, $y \in \{0,1\}^n$,
$w \in \mathbb{R}^d$, $b_0 \in \mathbb{R}$ (from LASSO fit),
$\boldsymbol{J_0}$: Youden's $\boldsymbol{J}$ for baseline model,
K: Vote magnitude,
rt: Relative tolerance to J required
Output: Rule Model (w^*, b^*)

1: Collect active features and sort by importance.

- Let $\mathcal{J} = \{j: w_j \neq 0\}$, $\ell = |\mathcal{J}|$
- Sort $\mathcal{J}$ by descending $|w_j|$ to get an index order $\pi = (\pi_1, \dots, \pi_\ell)$
- Define signs $s_j = sign(w_j) \in \{-1, +1\}$ for $j \in \mathcal{J}$.

2: Precompute per-feature signed contributions.

- Build: $M \in \{-1, +1\}^{n \times \ell}$ where $M_{i,k} = s_{\pi_k} X_{i,\pi_k}$
- Compute cumulative votes $V \in \mathbb{Z}^{n \times \ell}$ with $V_{i,k} = \sum_{t=1}^{k} M_{i,k}$ for $k = 1, \dots, \ell$
- So, $V_{i,k} = \#(agreements) - \#(disagreements)$ among the top-k.

3: Initialize incumbent (anytime state).

- Set $(w^*, b^*, k^*, J^*) \leftarrow (w, b_0, 0, -\infty)$
- $T \leftarrow (1 - rt) \cdot J_0$

4: Main scan over rule size k.

FOR k = 1 to ℓ

a. Construct the rule weight vector for top-k.
$$\widetilde{w}_j^{(k)} = \begin{cases} K \cdot s_j, & j \in \{\pi_1, \dots, \pi_k\} \\ 0, & otherwise \end{cases}$$

b. Choose intercept $b^{(k)}$ per policy.
$$b^{(k)} = log\left(\frac{p}{1-p}\right) - \sum_{t=1}^{k} K_{s_{\pi_t}} \mu_{\pi_t}$$
where:
$\mu_j = \frac{1}{n}\sum_i X_{i,j}$ and $p = \frac{1}{n}\sum_i y_i$

c. Update incumbent (for anytime).
if $J_k > J^*$:
update $(w^*, b^*, k^*, J^*) \leftarrow (\widetilde{w}^{(k)}, b^{(k)}, k, J_k)$

5: Adoption decision.

if $J^* > T$, adopt new coefficient set, otherwise $(w^*, b^*, k^*) \leftarrow (w, b_0, \ell)$.

6: Finalize outputs.

Return Rule Model (w^*, b^*) with k^* non-zeros, each $\pm K$ if the new coefficient set was adopted.

7: **end procedure**

3. Results

The custom solver consistently outperformed scikit-learn in both speed and scalability, running 1.5–2.3× faster on large rectified datasets while also using memory more efficiently through sparse binary feature representations [3]. The rule models showed no evidence of overfitting and in some cases generalized slightly better than their logistic counterparts, underscoring that compression not only enhances interpretability but *can* also act as an implicit regularizer (Fig. 1). These findings highlight the practicality of the approach for scaling to larger datasets without sacrificing performance.

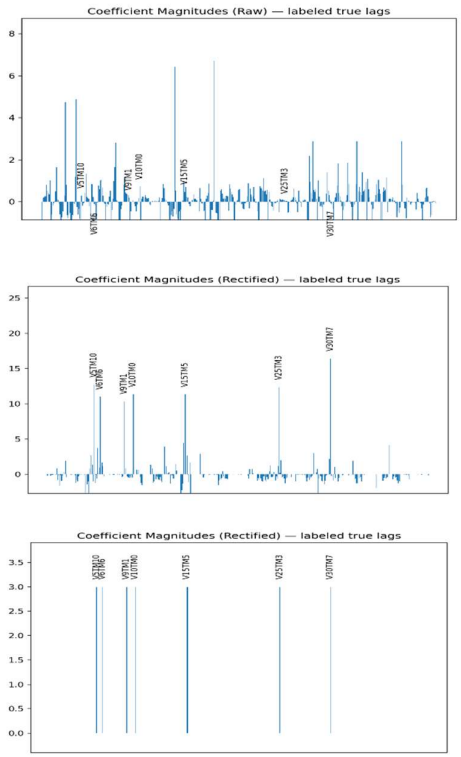

Fig. 1. Magnitudes for (top) L1-logistic on raw features, (middle) L1-logistic on rectified features (no polish), and (bottom) rectified L1-logistic after logic polishing. The polishing step collapses numerous tiny contributions into a handful of rules, achieving maximal sparsity without degrading J/AUC (adapted from [3]).

Taken together, these results demonstrate that the proposed rule-compression framework not only preserves predictive accuracy but also delivers scalable, interpretable models (Fig. 2), setting the stage for our concluding discussion of its broader implications.

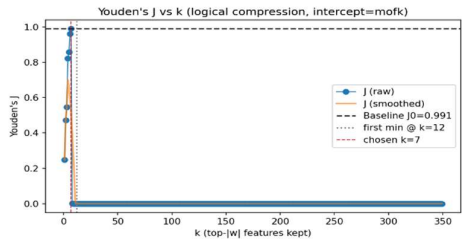

Fig. 2. Youden's J vs. rule count (k) curve shows how J on validation/test changes as the rule threshold requirement varies from 1 up to K (or total rules). The peak of the curve indicates the optimal "k chosen".

3.1. Quantifying Similarites

Across 36 paired simulation runs with differing seeds and data set sizes, all four AUC measures were above 0.97, differing by no more than ~0.03.

Pairwise paired t-tests [7] indicated statistically significant differences ($p < 0.001$ – see Table 1), but effect sizes were small (|Cohen's d| ≈ 1.0 for the largest contrast, typically < 0.3 for most).

Equivalence testing with a tolerance of ± 0.01 AUC confirmed that the base and rule variants are practically indistinguishable, implying that all methods converge to essentially the same predictive performance despite minor systematic offsets.

Table 1. Paired t-test AUC P-value Comparison.

Run	Bin'zed/ New	Rule Model	Bin'zed/ Scikit	Raw/ Scikit
Bin'zed/ New	NA	1.6E-6*	1.5E-6*	7.3E-8
Rule Model	1.6E-6*	NA	9.7E-10*	6.6E-7
Bin'zed/ Scikit	1.5E-6*	9.7E-10*	NA	7.7E-9
Raw/ Scikit	7.3E-8	6.6E-7	7.7E-9	NA

* These models also passed the "Two One-Sided Tests" (TOST) [8] criteria to consider these results statistically equivalent with 90% CI within ±0.01 AUC.

4. Conclusion

The proposed anytime logic-polishing framework converts sparse rectified logistic models into short rule sets that preserve the best attainable balance of sensitivity and specificity (via Youden's J) while retaining AUC performance. On synthetic longitudinal signals with known ground truth, we observed **up to ~50×** complexity reduction (hundreds of coefficients → single-digit rules) with equal or marginally better J and negligible AUC change, consistent full adoption, and 1.5–2.3× faster training using a custom solver for

binary inputs. These results suggest a practical pathway to deploy interpretable models for time-series monitoring in domains where clarity and verification by experts are essential.

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